

Exercise 10: Generalization

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Problem 1 (Rademacher Complexity):

In statistical learning theory, the *Rademacher complexity* is a measure of the complexity of a class of functions. It is used to bound the generalization error of learning algorithms.

Given a sample $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ drawn from some distribution $\mathcal{D}$, the empirical Rademacher complexity $\hat{\mathcal{R}}(\mathcal{F})$ of the hypothesis class $\mathcal{F}$ of binary classifier is defined as:

$$\hat{\mathcal{R}}(\mathcal{F}) = \mathbb{E}_{\boldsymbol{\sigma}} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(\mathbf{x}_i) \right],$$

where $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_n)$ are independent Rademacher random variables, i.e., $\mathbb{P}(\sigma_i = +1) = \mathbb{P}(\sigma_i = -1) = \frac{1}{2}$.

The Rademacher complexity quantifies the richness of the function class $\mathcal{F}$ by measuring how well functions in $\mathcal{F}$ can fit random noise. It is widely used to provide generalization bounds in machine learning: with probability at least $1 - \delta$ over sample draws, it holds that

$$\sup_{f \in \mathcal{F}} \left| \mathbb{E}_{x \sim \mathcal{D}}[f(x)] - \frac{1}{n} \sum_{i=1}^n f(x_i) \right| \leq 2\hat{\mathcal{R}}(\mathcal{F}) + 3\sqrt{\frac{2\log(1/\delta)}{n}}.$$

Now, consider the following questions about Rademacher complexity.

- a) For two hypothesis classes $\mathcal{F}, \mathcal{F}'$, prove that the Rademacher complexity of their sum satisfies the following inequality:

$$\hat{\mathcal{R}}(\mathcal{F} + \mathcal{F}') \leq \hat{\mathcal{R}}(\mathcal{F}) + \hat{\mathcal{R}}(\mathcal{F}').$$

- b) Consider an L -Lipschitz continuous function $\phi : \mathbb{R} \rightarrow \mathbb{R}$, i.e. for all $t, t' \in \mathbb{R}$,

$$|\phi(t) - \phi(t')| \leq L|t - t'|,$$

and define a new class $\mathcal{F}' := \{\phi \circ f : f \in \mathcal{F}\}$. Prove that the Rademacher complexity of $\mathcal{F}'$ is bounded as follows:

$$\hat{\mathcal{R}}(\mathcal{F}') \leq L\hat{\mathcal{R}}(\mathcal{F}).$$

- c) Let $\mathcal{F}$ be a class of real-valued functions and let $\ell(x) = \min(1, \max(0, x))$ be the (truncated) hinge loss. Show that for any $f \in \mathcal{F}$, with probability at least $1 - \delta$ over the sample draw, it holds that

$$\mathbb{E}[\ell(f(x), y)] \leq \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i) + 2\hat{\mathcal{R}}(\mathcal{F}) + 3\sqrt{\frac{2\log(1/\delta)}{n}}.$$

Problem 2 (PAC Bayes Bounds):

Let S denote a training set of size $|S| = m$,

$$S = \{(\mathbf{x}_i, y_i)\}_{i=1, \dots, m}, \text{ where } (\mathbf{x}_i, y_i) \in (\mathcal{X} \times \mathcal{Y}). \quad (1)$$

Let $\mathcal{M}$ denote the set of all probability measures on the data space $\mathbb{R}^k \times \{-1, 1\}$. We will assume that the training examples are i.i.d. samples from some $\mu \in \mathcal{M}$.

A parametric family of classifiers is a function $F : \mathbb{R}^d \times \mathbb{R}^k \rightarrow \{-1, 1\}$, where $f_{\mathbf{w}} := F(\mathbf{w}, \cdot) : \mathbb{R}^k \rightarrow \{-1, 1\}$ is the classifier indexed by the parameter $\mathbf{w} \in \mathbb{R}^d$. The hypothesis space induced by F is $\mathcal{F} = \{f_{\mathbf{w}} : \mathbf{w} \in \mathbb{R}^d\}$.

A randomized classifier is a distribution Q on $\mathbb{R}^d$. Informally, we will speak of distributions on $\mathcal{F}$ when we mean distributions on the underlying parametrization.

We are interested in the 0-1 loss $e_f : \mathbb{R}^k \times \{-1, 1\} \rightarrow \{0, 1\}$ defined by

$$e_f(\mathbf{x}, y) := \mathbb{I}[f(\mathbf{x}) \neq y].$$

We denote by $e_f^\mu := \mathbb{E}_\mu[e_f(\mathbf{x}, y)]$ the expected error and $e_f^S := \frac{1}{m} \sum_{i=1}^m e_f(\mathbf{x}_i, y_i)$ the empirical error w.r.t. sample $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^m$.

The main PAC-Bayesian theorem is the following:

Theorem 1 (McAllister 1999). *For fixed P and any Q , $\epsilon \in (0, 1)$, with probability at least $1 - \epsilon$ over sample sets $S \stackrel{i.i.d.}{\sim} \mu$:*

$$\mathbb{E}_Q[e_{f_{\mathbf{w}}}^\mu] - \mathbb{E}_Q[e_{f_{\mathbf{w}}}^S] \leq \sqrt{\frac{2}{|S|} \left[KL(Q||P) + \ln \left(\frac{2\sqrt{|S|}}{\epsilon} \right) \right]}. \quad (2)$$

By reviewing the lecture notes and/or reading the paper by Dziugaite and Roy “Computing nonvacuous generalization bounds for deep (stochastic) neural networks with many more parameters than training data” <https://arxiv.org/abs/1703.11008>, answer the following questions:

- What is a “non-vacuous” generalization bound, and when is a generalization bound considered “non-vacuous” (in contrast to when it is considered to be “vacuous”)?
- In the proof of the McAllister Theorem, we need the condition that Q is absolutely continuous with respect to P . What does this mean, and why is it needed?
- Describe in simple words what the McAllister theorem tells us. What is the role of P and Q ? Why is the PAC Bayes bound considered to be a data-dependent generalization bound (e.g. in contrast to data-independent VC bounds)?
- How is the PAC Bayes bound used for deep neural networks (DNNs) in practice? What is a stochastic neural network, and how is it implemented in practice?
- The KL divergence is minimized when $Q = P$. To get bounds on a deterministic classifier, one might imagine choosing Q to be a Dirac-delta distribution which is non-zero for only one weight vector. Would that work? Explain.